<u>Program - 4:</u> Design a profile card UI using Container, Row, Column, Image, and Text widgets.

<u>Code :</u>

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    home: Scaffold(
        appBar: AppBar(title: Text('Profile Card')),
        body: Center(
          child: Container(
            width: 320,
            padding:  EdgeInsets.all(15),
            decoration:   BoxDecoration(
            color: Colors.grey.shade200,
              borderRadius: BorderRadius.circular(12),
              border: Border.all(color: Colors.blue),
            ),
            child: Column(
              mainAxisSize: MainAxisSize.min,
              children: [
                Container(
                  width: 100,
                  height: 150,
                  decoration: BoxDecoration(
```

```
            border: Border.all(
              color: const Color.fromARGB(255, 8, 8, 8),
            ),
          ),
          child: Image.asset('assets/1.jpg', fit: BoxFit.cover),
        ),
        SizedBox(height: 15),
        Text(
          'Name: Kayasth Harsh,
          style: TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
        ),
        SizedBox(height: 6),
        Text('Course : BCA'),

        SizedBox(height:6),
        Text('College: Sarvajanik College of Commerce'),
      ],
    ),
  ),
),
),
);
}
}
```

Output :

10:49
Profile Card
Name: Kayasth Harsh
Course: BCA
College: Sarvajanik Coll...

<u>Program - 5:</u> Create a simple calculator UI layout using buttons and text fields (without functionality).

<u>Code :</u>

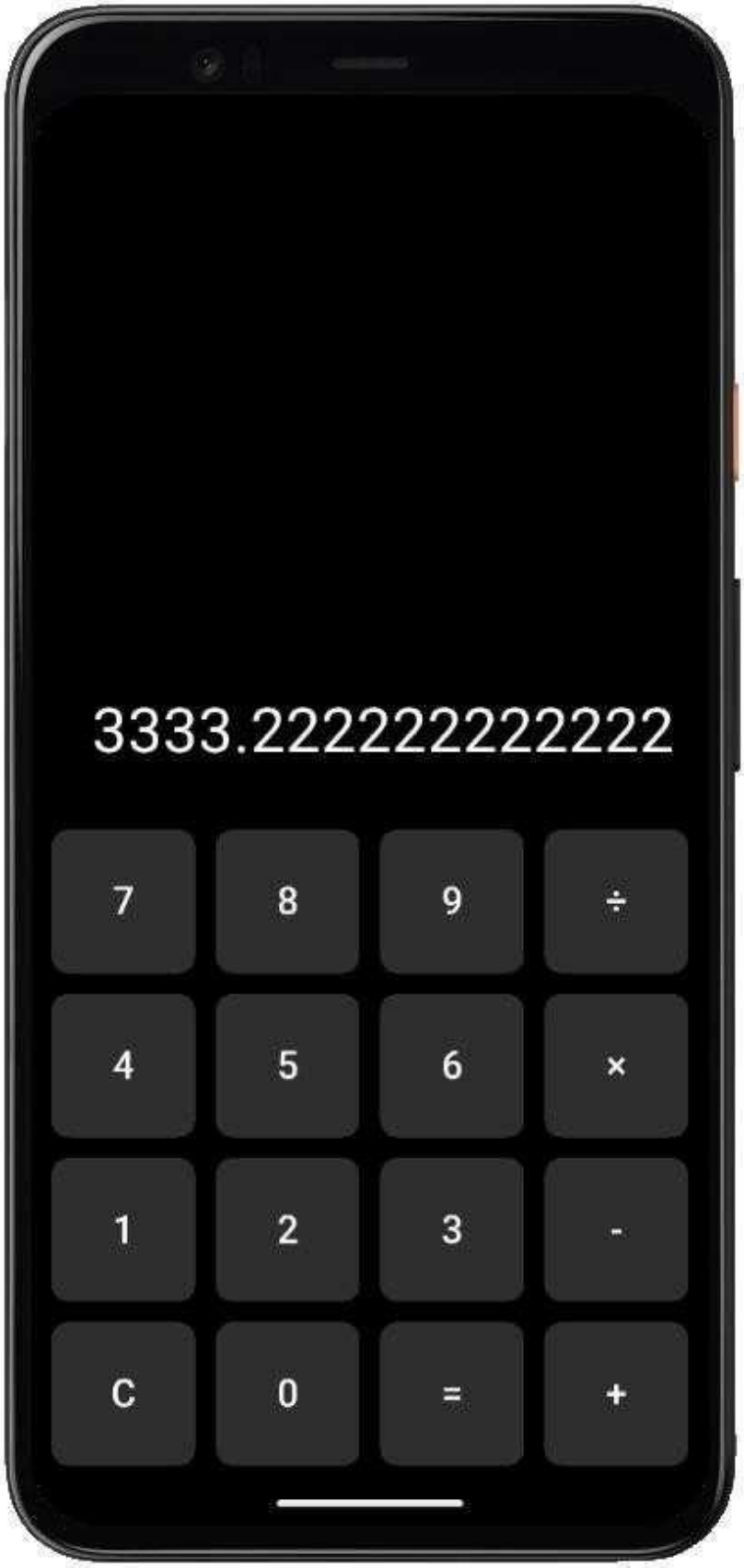
```
import 'package:flutter/material.dart';
void main() {
  runApp(const CalculatorApp());
}
class CalculatorApp extends StatelessWidget {
  const CalculatorApp({super.key});
  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
    debugShowCheckedModeBanner: false,
    home: CalculatorUI(),
    );
  }
}
class CalculatorUI extends StatefulWidget {
  const CalculatorUI({super.key});
  @override
  State<CalculatorUI> createState() => _CalculatorUIState();
}
class _CalculatorUIState extends State<CalculatorUI> {
  String display = "";
  double num1 = 0;
  double num2 = 0;
  String operator = "";
  void onButtonPress(String value) {
    setState(() {
      if (value == "+" || value == "-" || value == "×" || value == "÷") {
```

```
      num1 = double.parse(display);
      operator = value;
      display = "";
    } else if (value == "=") {
      num2 = double.parse(display);
      if (operator == "+") {
        display = (num1 + num2).toString();
      } else if (operator == "-") {
        display = (num1 - num2).toString();
      } else if (operator == "×") {
        display = (num1 * num2).toString();
      } else if (operator == "÷") {
        display = (num1 / num2).toString();
      }
    } else if (value == "C") {
      display = "";
      num1 = 0;
      num2 = 0;
      operator = "";
    } else {
      display += value;
    }
  });
}
Widget calcButton(String text) {
  return ElevatedButton(
    onPressed: () => onButtonPress(text),
    style: ElevatedButton.styleFrom(
    backgroundColor: Colors.grey[850],
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(10)),
    ),
    child: Text(
```

```
      text,
      style: const TextStyle(fontSize: 22, color: Colors.white),
    ),
  );
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: Colors.black,
    body: SafeArea(
      child: Column(
        children: [
        Expanded(
child: Align(
  alignment: Alignment.bottomRight,
  child: Padding(
    padding: const EdgeInsets.all(20),
    child: Text(
      display,
      style: const TextStyle(fontSize: 36, color: Colors.white),
    ),
  ),
),
),
          Expanded(
            child: GridView.count(
              crossAxisCount: 4,
              padding: const EdgeInsets.all(12),
              mainAxisSpacing: 12,
              crossAxisSpacing: 12,
              children: [
              calcButton("7"),
```

```
            calcButton("8"),
            calcButton("9"),
            calcButton("÷"),
            calcButton("4"),
            calcButton("5"),
            calcButton("6"),
            calcButton("×"),
            calcButton("1"),
            calcButton("2"),
            calcButton("3"),
            calcButton("-"),
            calcButton("C"),
            calcButton("0"),
            calcButton("="),
            calcButton("+"),
          ],
        ),
      ),
    ],
  ),
),
);
}
}
```

Output :



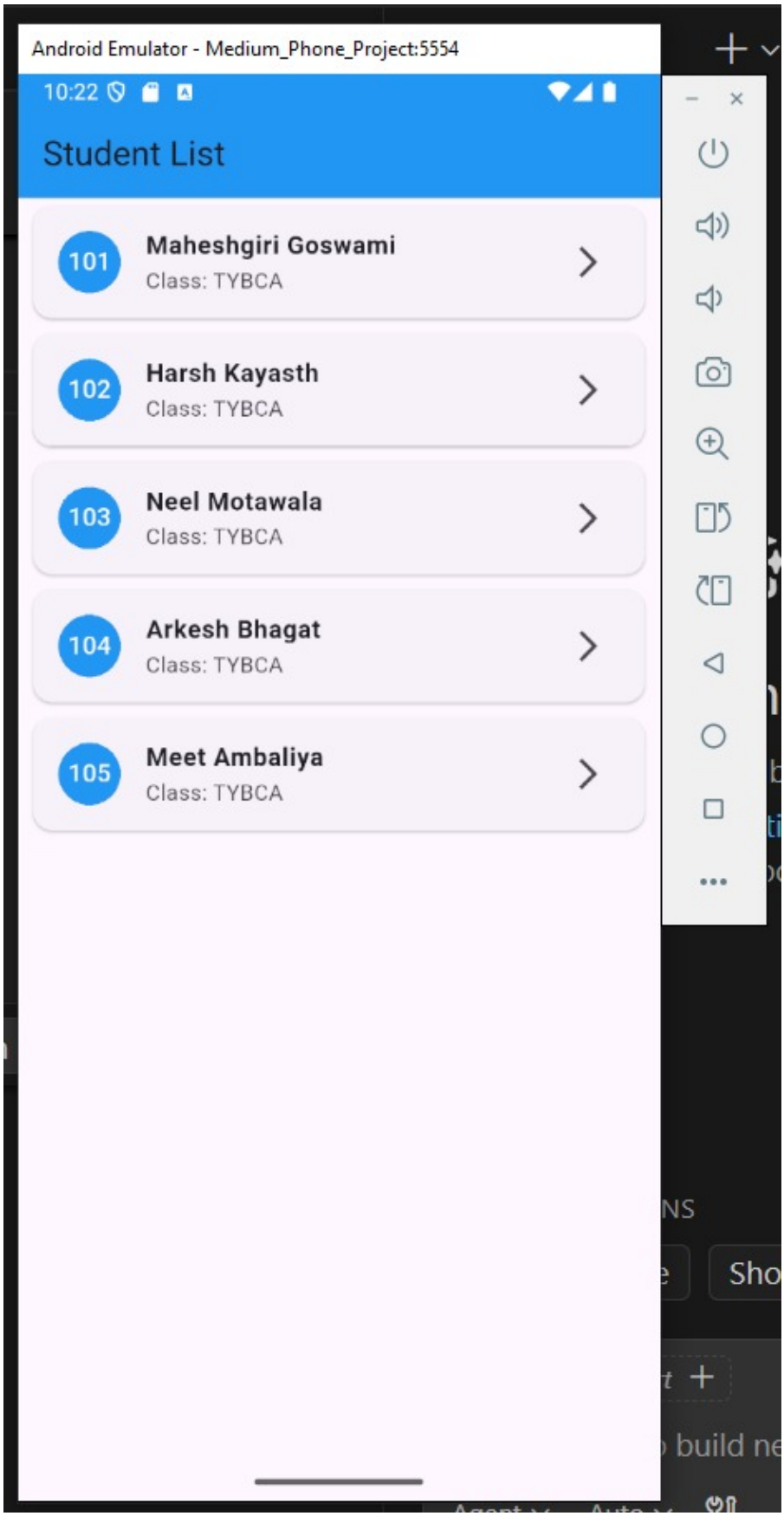
Program - 6: Build a student list page using ListView and ListTitle widgets.

Code :

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    home: StudentListPage(),
    );
  }
}
class StudentListPage extends StatelessWidget {
  final List<Map<String, String>> students = [

  {"roll": "101", "name": "Maheshgiri Goswami", "class": "TYBCA"},
      {"roll": "102", "name": "Harsh Kayasth", "class": "TYBCA"},
      {"roll": "103", "name": "Neel Motawala", "class": "TYBCA"},
      {"roll": "104", "name": "Arkesh Bhagat", "class": "TYBCA"},
      {"roll": "105", "name": "Meet Ambaliya", "class": "TYBCA"},];
  StudentListPage({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Student List"),
```

```
    backgroundColor: Colors.blue,
  ),
  body: ListView.builder(
    itemCount: students.length,
    itemBuilder: (context, index) {
    return Card(
      margin: const EdgeInsets.symmetric(horizontal: 10, vertical: 5),
      child: ListTile(
        leading: CircleAvatar(
          backgroundColor: Colors.blue,
          child: Text(
          students[index]['roll']!,
            style: const TextStyle(color: Colors.white),
          ),
        ),
        title: Text(
          students[index]['name']!,
          style: const TextStyle(fontWeight: FontWeight.bold),
        ),
        subtitle: Text("Class: ${students[index]['class']}"),
        trailing: const Icon(Icons.arrow_forward_ios),
        onTap: () {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(
            content: Text("Selected: ${students[index]['name']}"),
          ),
        );
        },
      ),
    );
  },
),  ); }}
```

Output :


Android Emulator - Medium_Phone_Project:5554
10:22
Student List
101
Maheshgiri Goswami
Class: TYBCA
102
Harsh Kayasth
Class: TYBCA
103
Neel Motawala
Class: TYBCA
104
Arkesh Bhagat
Class: TYBCA
105
Meet Ambaliya
Class: TYBCA

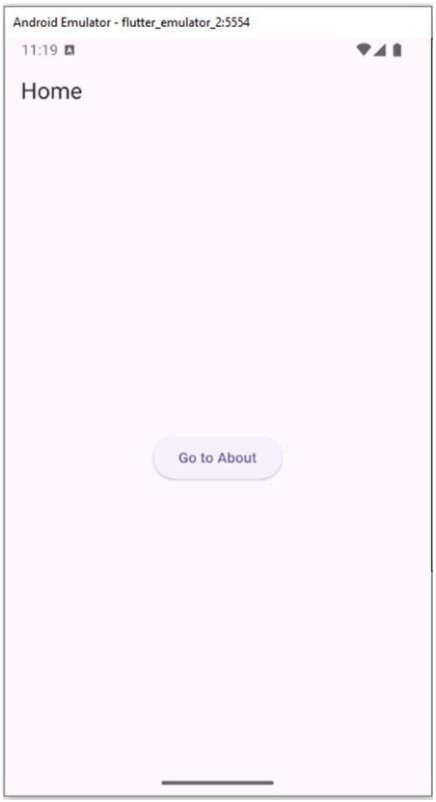
Program - 7: Create a multiscreen Navigation (E.g. Home Screen => About Screen => Contact Screen).

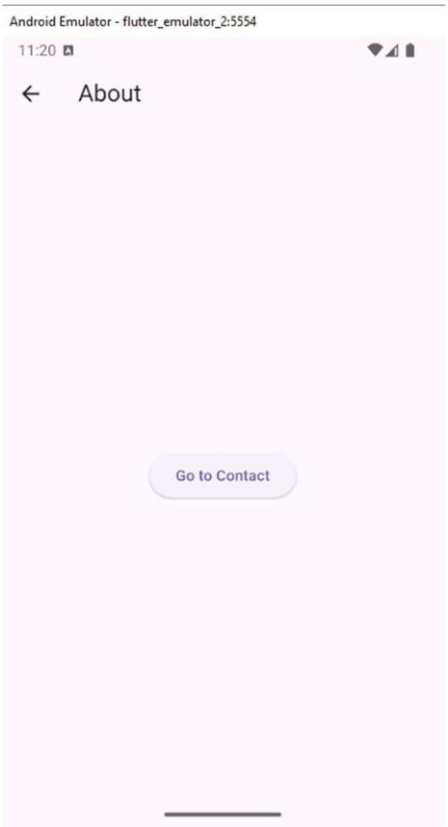
Code:

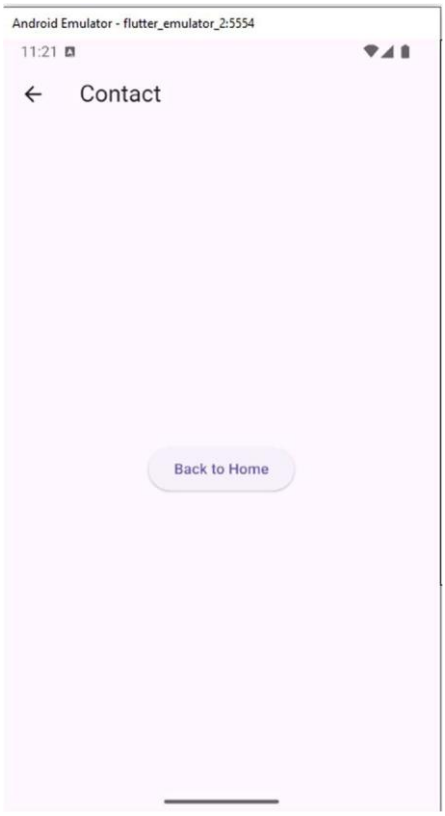
```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Multi Screen Navigation',
      home: const HomeScreen(),
    );
  }
}
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Home')),
      body: Center(
        child: ElevatedButton(
          child: const Text('Go to About'),
          onPressed: () {
            Navigator.push(
              context,
              MaterialPageRoute(builder: (context) => const AboutScreen()),
            );
          },
        ),
      ),
    );
  }
}
```

```
class AboutScreen extends StatelessWidget {
  const AboutScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('About')),
      body: Center(
        child: ElevatedButton(
          child: const Text('Go to Contact'),
          onPressed: () {
            Navigator.push(
              context,
              MaterialPageRoute(builder: (context) => const ContactScreen()),
            );
          },
        ),
      ),
    );
  }
}
class ContactScreen extends StatelessWidget {
  const ContactScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Contact')),
      body: Center(
        child: ElevatedButton(
          child: const Text('Back to Home'),
          onPressed: () {
          Navigator.pop(context);
          },
        ),
      ),
    );
  }
}
```

Output :


Android Emulator - flutter_emulator_2:5554
11:19
Home
Go to About


Android Emulator - flutter_emulator_2:5554
11:20
About
Go to Contact


Android Emulator - flutter_emulator_2:5554
11:21
Contact
Back to Home

<u>Program - 8:</u> Make a login form with username and password input field , validate input , and navigate to the home page.

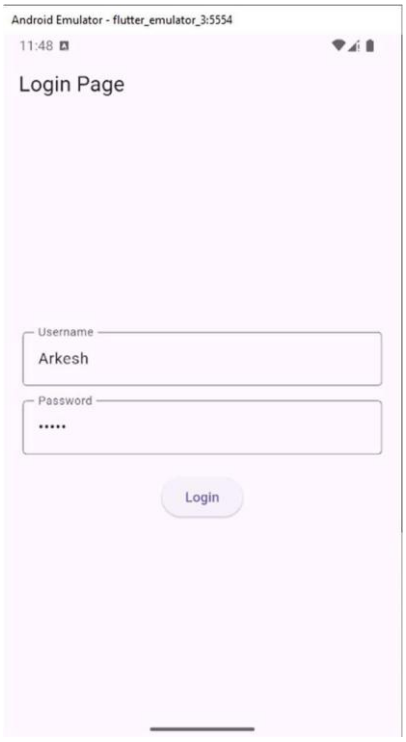
<u>Code :</u>

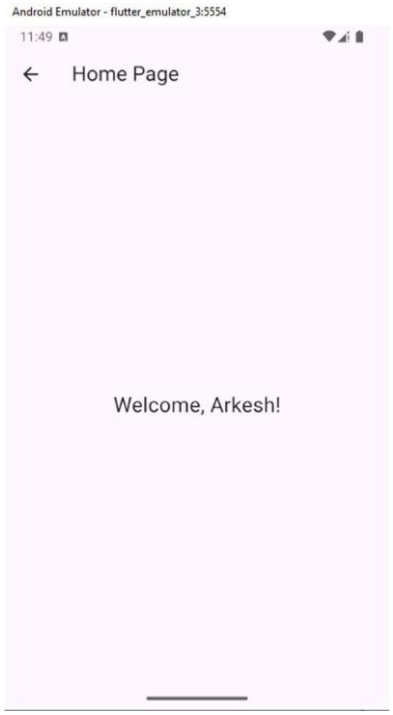
```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
// Main App
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(debugShowCheckedModeBanner: false, home: LoginPage());
  }
}
// Login Page
class LoginPage extends StatefulWidget {
  @override
  State<LoginPage> createState() => _LoginPageState();
}
class _LoginPageState extends State<LoginPage> {
  // Controllers to get text from TextField
  TextEditingController usernameController = TextEditingController();
  TextEditingController passwordController = TextEditingController();
  // Method for login validation
  void login() {
    String username = usernameController.text;
    String password = passwordController.text;
    if (username.isEmpty || password.isEmpty) {
      // Show error message
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text("Please enter username and password")),
      );
    } else {
      // Navigate to Home Page
      Navigator.push(
        context,
        MaterialPageRoute(builder: (context) => HomePage(username: username)),
      );
    }
  }
```

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Login Page")),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            // Username TextField
            TextField(
              controller: usernameController,
              decoration: InputDecoration(
              labelText: "Username",
                border: OutlineInputBorder(),
              ),
            ),
            SizedBox(height: 15),
            // Password TextField
            TextField(
              controller: passwordController,
              obscureText: true,
              decoration: InputDecoration(
                labelText: "Password",
                border: OutlineInputBorder(),
              ),
            ),
            SizedBox(height: 20),

            // Login Button
            ElevatedButton(onPressed: login, child: Text("Login")),
          ],
        ),
      ),
    );
  }
}

// Home Page
class HomePage extends StatelessWidget {
```

```
  final String username;
  HomePage({required this.username});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Home Page")),
      body: Center(
        child: Text("Welcome, $username!", style: TextStyle(fontSize: 22)),
      ),
    );
  }
}
```

Output :





Program - 9: Store and retrieve simple Key-value Pairs (e.g. username) using Shared Preferences.

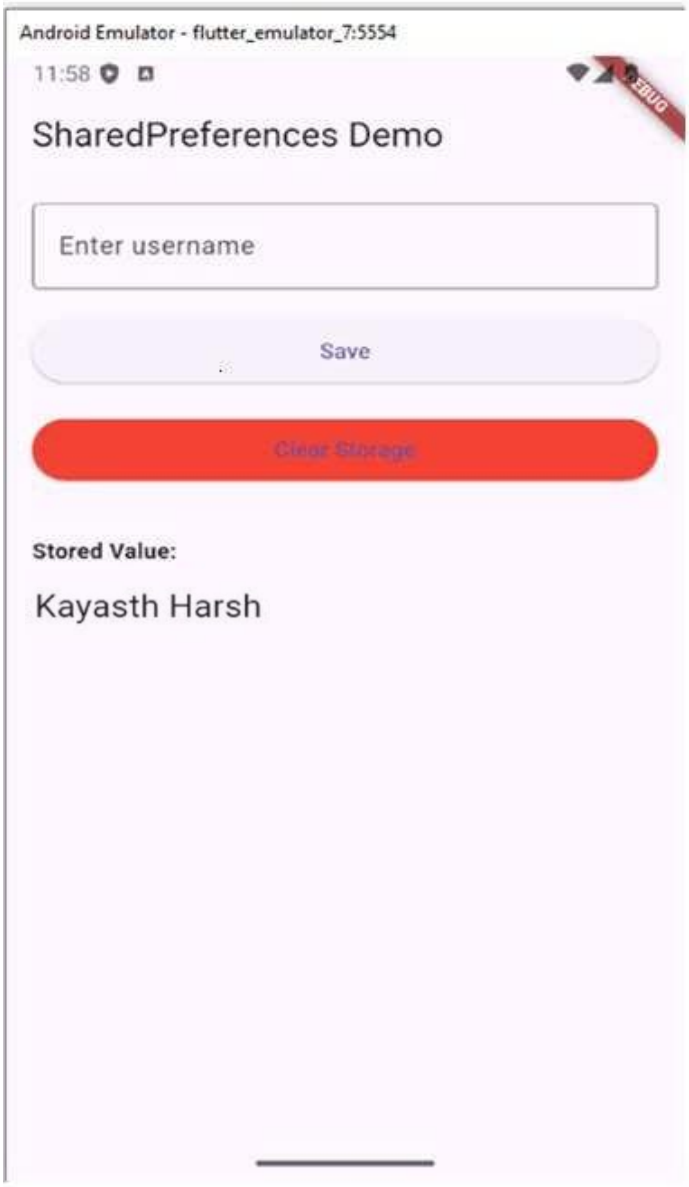
Code :

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'SharedPrefs Demo',
      theme: ThemeData(primarySwatch: Colors.blue),
      home: StorageScreen(),
    );
  }
}
class StorageScreen extends StatefulWidget {
  @override
  _StorageScreenState createState() => _StorageScreenState();
}
class _StorageScreenState extends State<StorageScreen> {
  final TextEditingController _controller = TextEditingController();
  String _storedValue = 'Nothing stored yet';

  @override
  void initState() {
    super.initState();
    _loadData();
  }
  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
  Future<void> _loadData() async {
    final prefs = await SharedPreferences.getInstance();
    setState(() {
      _storedValue = prefs.getString('username') ?? 'Nothing stored yet';
    });
```

```
  }

  Future<void> _saveData() async {
    final prefs = await SharedPreferences.getInstance();
    await prefs.setString('username', _controller.text);
    _controller.clear();
    await _loadData();
  }
  Future<void> _clearData() async {
    final prefs = await SharedPreferences.getInstance();
    await prefs.remove('username');
    await _loadData();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('SharedPreferences Demo')),
      body: Padding(
        padding: const EdgeInsets.all(16.0),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.stretch,
          children: [
            TextField(
              controller: _controller,
              decoration: InputDecoration(
              labelText: 'Enter username',
              border: OutlineInputBorder(),
              ),
            ),
            SizedBox(height: 16),
            ElevatedButton(onPressed: _saveData, child: Text('Save')),
            SizedBox(height: 16),
            ElevatedButton(
              onPressed: _clearData,
              style: ElevatedButton.styleFrom(backgroundColor: Colors.red),
              child: Text('Clear Storage'),
            ),
            SizedBox(height: 32),
            Text(
              'Stored Value:',
              style: TextStyle(fontWeight: FontWeight.bold),
```

```
          ),
          SizedBox(height: 8),
          Text(_storedValue, style: TextStyle(fontSize: 18)),
        ],
      ),
    ),
  );
 }
}
```

Output :



Program - 10: Fetch and display JSON data from a public API (e.g.,list of posts or users) using the http package.

Code :

Import the shared preferences package:

```
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'dart:convert'; // For JSON decoding
```

Open the pubspec.yaml file and add the below code :

```
dependencies:
 flutter:
 sdk: flutter
 shared_preferences: ^1.3.0
```

Then run this command in terminal :

```
flutter pub get
```

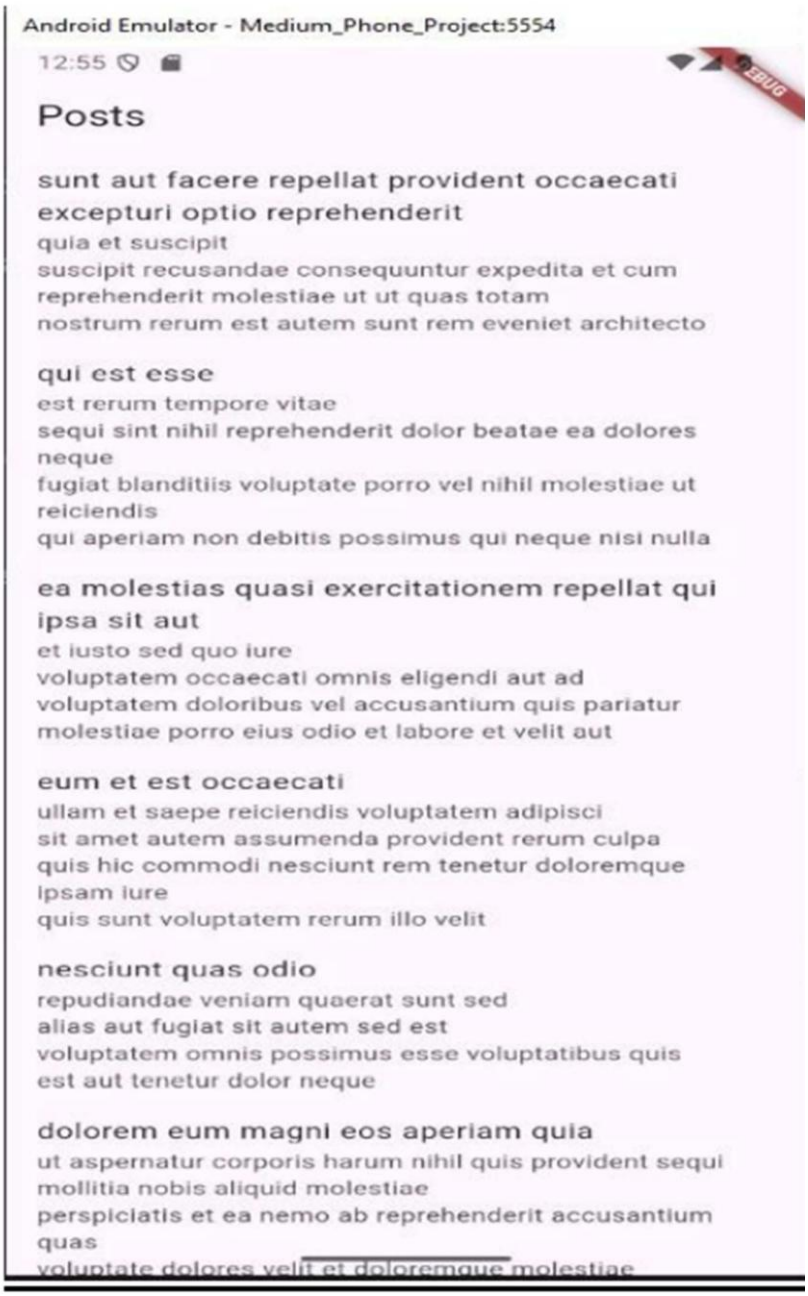
Main.dart code :

```
void main() {
 runApp(MyApp());
}
class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
 return MaterialApp(
 title: 'JSON Fetch Demo',
 theme: ThemeData(primarySwatch: Colors.blue),
 home: PostsScreen(),
 );
 }
}
class PostsScreen extends StatefulWidget {
 @override
 _PostsScreenState createState() => _PostsScreenState();
}
class _PostsScreenState extends State<PostsScreen> {
 List<dynamic> posts = [];
 bool isLoading = true;
 @override
 void initState() {
 super.initState();
 fetchPosts();
 }
 Future<void> fetchPosts() async {
```

```
try {
final response = await http.get(
Uri.parse('https://jsonplaceholder.typicode.com/posts'),
);
if (response.statusCode == 200) {
setState(() {
posts = jsonDecode(response.body);
isLoading = false;
});
}
else {
throw Exception('Failed to load posts');
}
}
catch (e) {
setState(() {
isLoading = false;
});
ScaffoldMessenger.of(context).showSnackBar(
SnackBar(content: Text('Error: $e')),
);
}
}
@override
Widget build(BuildContext context) {
return Scaffold(
appBar: AppBar(title: Text('Posts')),
body: isLoading
? Center(child: CircularProgressIndicator())
: ListView.builder(
itemCount: posts.length,
itemBuilder: (context, index) {
final post = posts[index];
return Card(
margin: EdgeInsets.all(8),
child: ListTile(
title: Text(post['title']),
subtitle: Text(post['body']),
leading: CircleAvatar(
child: Text((index + 1).toString()),
),
```

```
 ),
 );
 },
 ),
 );
 }
}
```

Output :


Android Emulator - Medium_Phone_Project:5554
12:55
DEBUG
Posts
sunt aut facere repellat provident occaecati excepturi optio reprehenderit
quia et suscipit
suscipit recusandae consequuntur expedita et cum reprehenderit molestiae ut ut quas totam
nostrum rerum est autem sunt rem eveniet architecto
qui est esse
est rerum tempore vitae
sequi sint nihil reprehenderit dolor beatae ea dolores neque
fugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis
qui aperiam non debitis possimus qui neque nisi nulla
ea molestias quasi exercitationem repellat qui ipsa sit aut
et iusto sed quo iure
voluptatem occaecati omnis eligendi aut ad
voluptatem doloribus vel accusantium quis pariatur
molestiae porro eius odio et labore et velit aut
eum et est occaecati
ullam et saepe reiciendis voluptatem adipisci
sit amet autem assumenda provident rerum culpa
quis hic commodi nesciunt rem tenetur doloremque ipsam iure
quis sunt voluptatem rerum illo velit
nesciunt quas odio
repudiandae veniam quaerat sunt sed
alias aut fugiat sit autem sed est
voluptatem omnis possimus esse voluptatibus quis
est aut tenetur dolor neque
dolorem eum magni eos aperiam quia
ut aspernatur corporis harum nihil quis provident sequi
mollitia nobis aliquid molestiae
perspiciatis et ea nemo ab reprehenderit accusantium quas
voluptate dolores velit et doloremque molestiae

Program - 11: take a photo using the device camera and displays it

Code :

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:image_picker/image_picker.dart';
void main() {
 runApp(MyApp());
}
class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
 return MaterialApp(
 title: 'Camera App',
 home: CameraPage(),
 );
 }
}
class CameraPage extends StatefulWidget {
 @override
 _CameraPageState createState() => _CameraPageState();
}
class _CameraPageState extends State<CameraPage> {
 File _image;
 final picker = ImagePicker();
 Future getImage() async {
 final pickedFile = await picker.getImage(source: ImageSource.camera);
 setState(() {
 if (pickedFile != null) {
 _image = File(pickedFile.path);
 } else {
 print('No image selected.');
 }
 });
 }
 @override
 Widget build(BuildContext context) {
 return Scaffold(
 appBar: AppBar(
 title: Text('Camera App'),
 ),
 body: Center(
```

```
 child: _image == null
 ? Text('No image selected.')
 : Image.file(_image),
 ),
 floatingActionButton: FloatingActionButton(
 onPressed: getImage,
 tooltip: 'Take Photo',
 child: Icon(Icons.camera_alt),
 ),
 );
 }
}
```

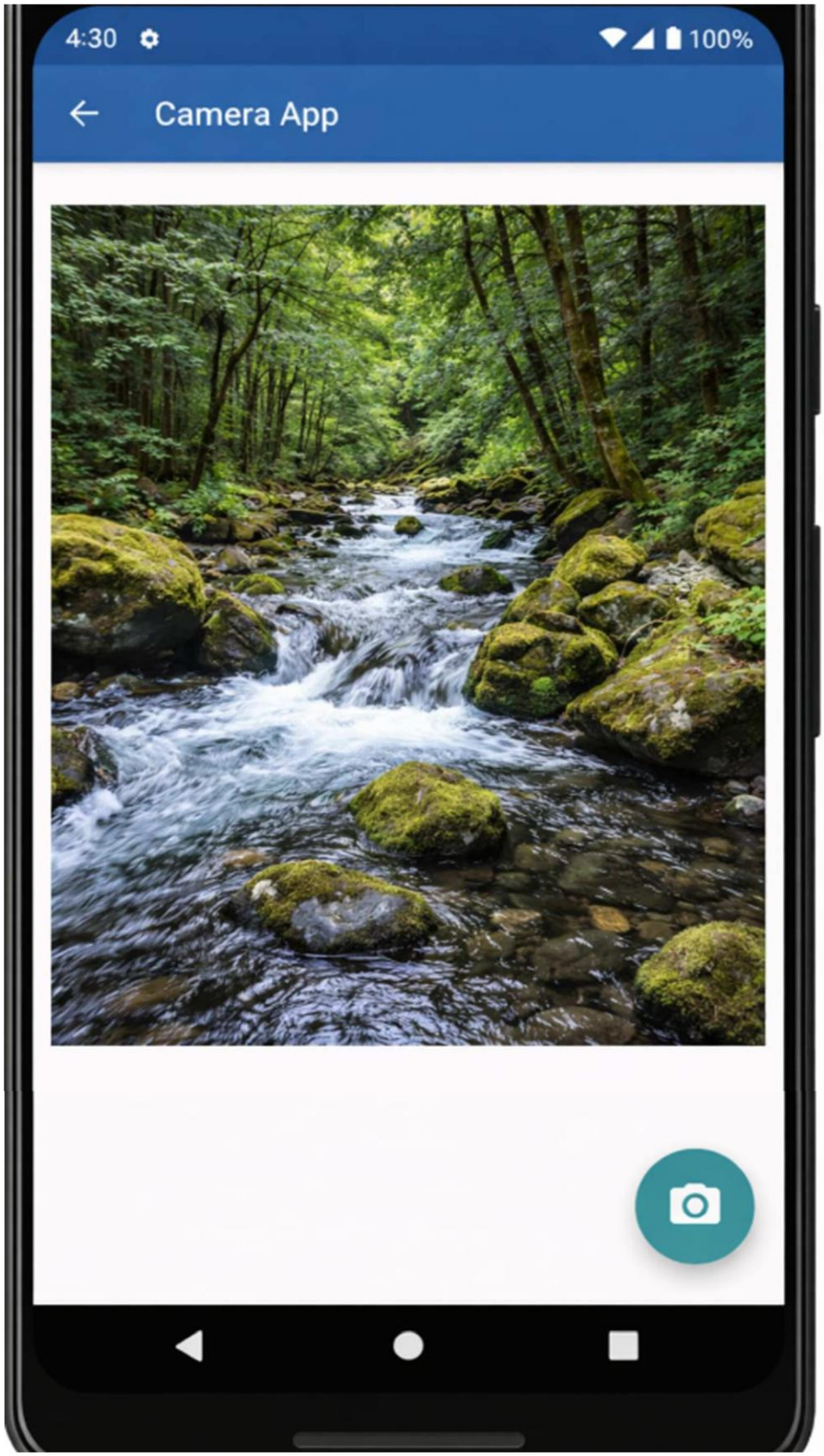
<u>the changes needed in the config files:</u>

- pubspec.yaml:

- You need to add the image_picker package to your dependencies:

```
dependencies:
    flutter:
    sdk: flutter
    image_picker: ^0.8.0
```

Output :



Program - 12: Connect a Flutter app with Firebase Authentication for Email/Password login.

Code :

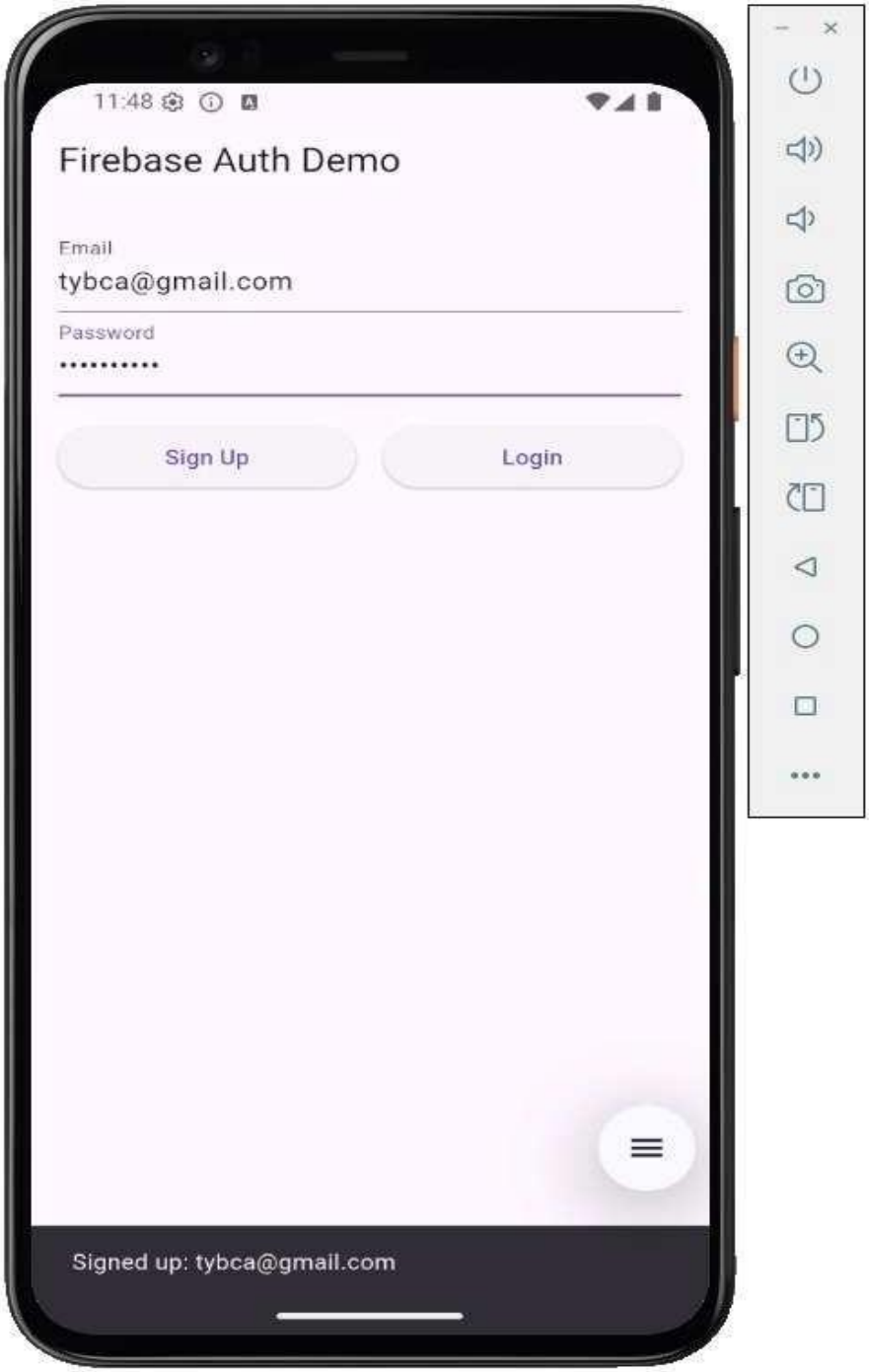
```
dependencies:
  flutter:
   sdk: flutter
  firebase_core: ^4.5.0
  firebase_auth: ^6.2.0
  flutter_secure_storage: ^9.0.0

import 'package:flutter/material.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:firebase_core/firebase_core.dart';

Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
    debugShowCheckedModeBanner: false,
    home: AuthDemoScreen(),
    );
  }
}
class AuthDemoScreen extends StatefulWidget {
  const AuthDemoScreen({super.key});
  @override
  State<AuthDemoScreen> createState() => _AuthDemoScreenState();
}
class _AuthDemoScreenState extends State<AuthDemoScreen> {
  final _emailController = TextEditingController();
  final _passwordController = TextEditingController();
  final FirebaseAuth _auth = FirebaseAuth.instance;
```

```
Future<void> _signup() async {
  try {
    UserCredential userCredential =
        await _auth.createUserWithEmailAndPassword(
      email: _emailController.text.trim(),
      password: _passwordController.text,
    );

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
          content: Text('Signed up: ${userCredential.user?.email}')),
    );
  } on FirebaseAuthException catch (e) {
    ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(content: Text('Signup error: ${e.message}')),
    );
  }
}

Future<void> _login() async {
  try {
    UserCredential userCredential =
        await _auth.signInWithEmailAndPassword(
      email: _emailController.text.trim(),
      password: _passwordController.text,
    );

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
          content: Text('Logged in: ${userCredential.user?.email}')),
    );
  } on FirebaseAuthException catch (e) {
    ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(content: Text('Login error: ${e.message}')),
    );
  }
}

@override
void dispose() {
```

```
  _emailController.dispose();
  _passwordController.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Firebase Auth Demo'),
    ),
    body: Padding(
      padding: const EdgeInsets.all(16),
      child: Column(
        children: [
          TextField(
            controller: _emailController,
            decoration: const InputDecoration(labelText: 'Email'),
            keyboardType: TextInputType.emailAddress,
          ),
          TextField(
            controller: _passwordController,
            decoration: const InputDecoration(labelText: 'Password'),
            obscureText: true,
          ),
          const SizedBox(height: 16),
          Row(
            children: [
              Expanded(
                child: ElevatedButton(
                  onPressed: _signup,
                  child: const Text('Sign Up'),
                ),
              ),
              const SizedBox(width: 16),
              Expanded(
                child: ElevatedButton(
                  onPressed: _login,
                  child: const Text('Login'),
                ),
              ),
```

```
            ],
          ),
        ],
      ),
    ),
  );
 }
}
```

Output :


11:48
Firebase Auth Demo
Email
tybca@gmail.com
Password
Sign Up
Login
Signed up: tybca@gmail.com

<u>Program - 13:</u> Build a To-Do List app using Firebase Cloud Firestore for storing and retrieving tasks.

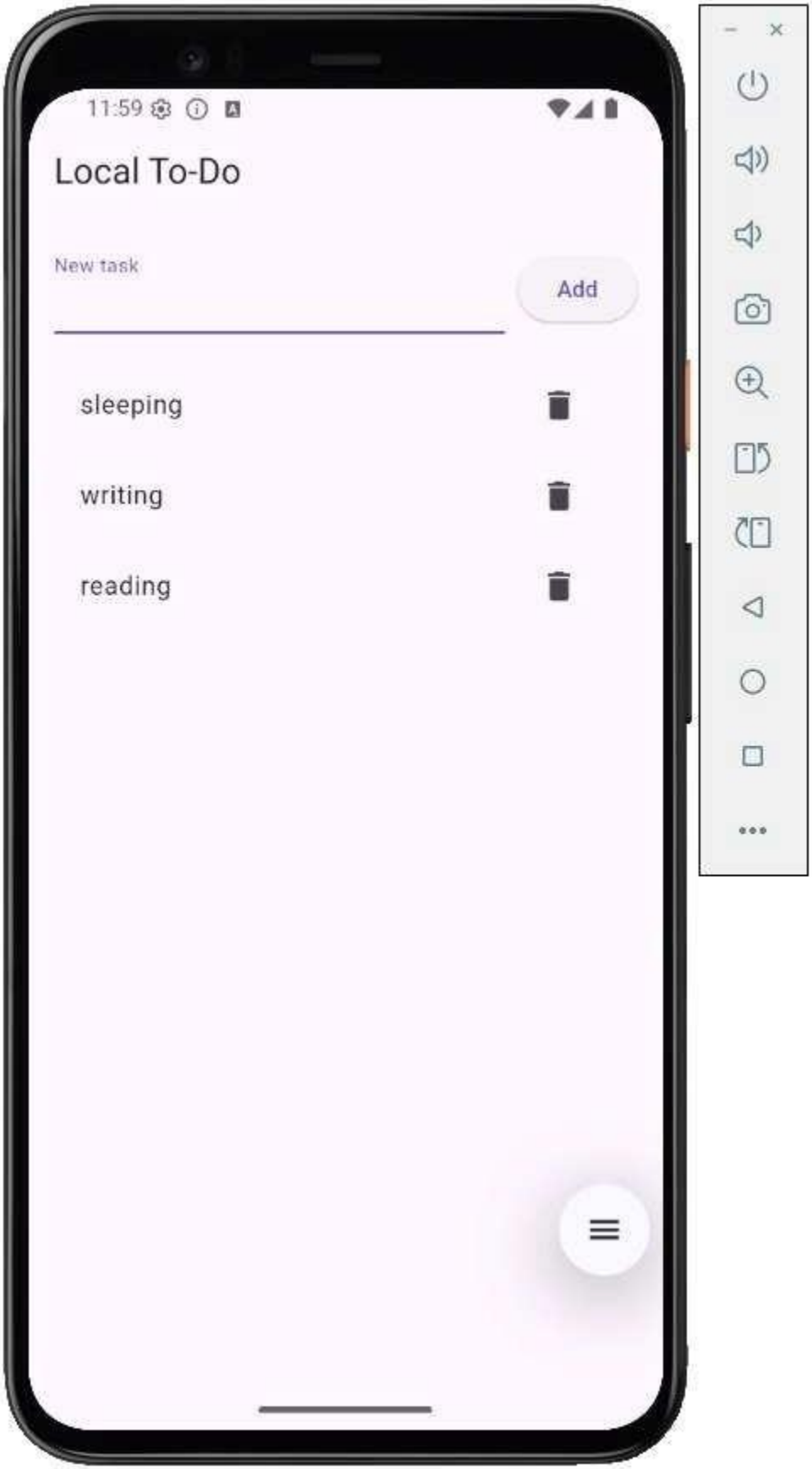
<u>Code :</u>

```
dependencies:
  flutter:
   sdk: flutter
  firebase_core: ^4.5.0
  firebase_auth: ^6.2.0
  flutter_secure_storage: ^9.0.0

import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';
import 'dart:convert';
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Local To-Do',
      home: TodoScreen(),
    );
  }
}
class TodoScreen extends StatefulWidget {
  const TodoScreen({super.key});
  @override
  State<TodoScreen> createState() => _TodoScreenState();
}
class _TodoScreenState extends State<TodoScreen> {
  final TextEditingController _controller = TextEditingController();
  List<String> _tasks = [];
  @override
  void initState() {
    super.initState();
    _loadTasks();
```

```
  }
  Future<void> _loadTasks() async {
    final prefs = await SharedPreferences.getInstance();
    final String? tasksJson = prefs.getString('tasks');
    if (tasksJson != null) {
      setState(() {
        _tasks = List<String>.from(json.decode(tasksJson));
      });
    }
  }
  Future<void> _saveTasks() async {
    final prefs = await SharedPreferences.getInstance();
    await prefs.setString('tasks', json.encode(_tasks));
  }
  Future<void> _addTask() async {
    final text = _controller.text.trim();
    if (text.isEmpty) return;
    setState(() {
      _tasks.insert(0, text);
    });
    await _saveTasks();
    _controller.clear();
  }
  Future<void> _deleteTask(int index) async {
    setState(() {
      _tasks.removeAt(index);
    });
    await _saveTasks();
  }
  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Local To-Do')),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
```

```
        children: [
          Row(
          children: [
            Expanded(
              child: TextField(
                controller: _controller,
                decoration:
                    const InputDecoration(labelText: 'New task'),
              ),
            ),
            const SizedBox(width: 8),
            ElevatedButton(
            onPressed: _addTask,
            child: const Text('Add'),
            ),
          ],
          ),
          const SizedBox(height: 16),
          Expanded(
            child: _tasks.isEmpty
                ? const Center(child: Text('No tasks yet.'))
                : ListView.builder(
                    itemCount: _tasks.length,
                    itemBuilder: (context, index) {
                      return ListTile(
                        title: Text(_tasks[index]),
                        trailing: IconButton(
                          icon: const Icon(Icons.delete),
                          onPressed: () => _deleteTask(index),
                        ),
                      );
                    },
                  ),
          ),
        ],
      ),
    ),
  );
 }
}
```

Output :



<u>Program - 14:</u> Build and generate an APK for your app and install it on a mobile device.

<u>Code :</u>

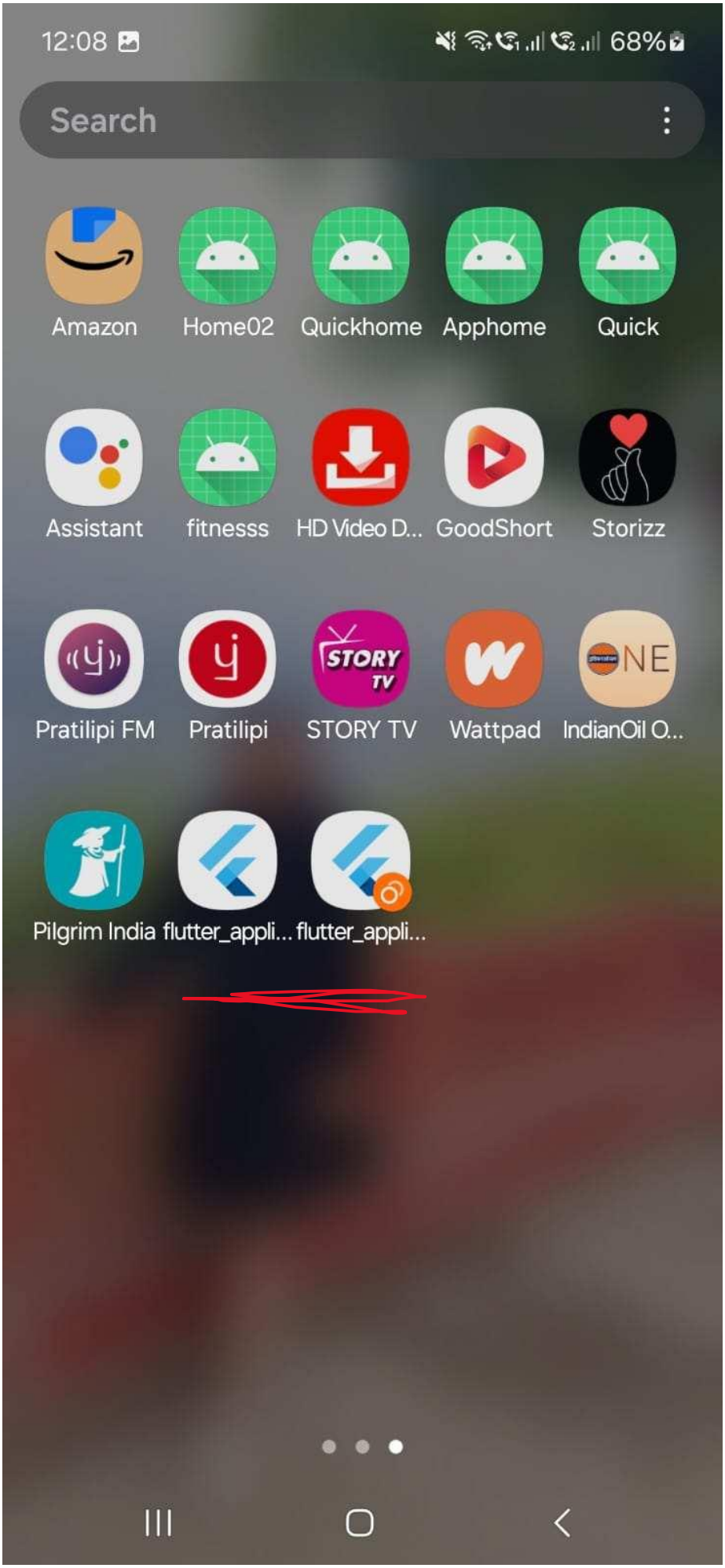
```
dependencies:
  flutter:
   sdk: flutter
  shared_preferences: ^2.5.4

import 'package:flutter/material.dart';
void main() {
runApp(MyApp());
}
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
debugShowCheckedModeBanner: false,
home: Scaffold(
appBar: AppBar(
title: Text('Personal Details'),
),
body: Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.center,
children: [
Text(
'Name: Bina',
style: TextStyle(fontSize: 20),
),
SizedBox(height: 10),
Text(
'Course: BCA',
style: TextStyle(fontSize: 20),
),
SizedBox(height: 10),
Text(
'College: Sarvajanik College of Commerce & Computer Application',
style: TextStyle(fontSize: 20),
textAlign: TextAlign.center,
),
```

```
],
),
),
),
);
}
}
```

Output :

```
Chrome (web)                • chrome        • web-javascript • Google Chrome 145.0.7
Edge (web)                  • edge          • web-javascript • Microsoft Edge 145.0.
[1]: SM A356E (RZCX7049DDM)
[2]: sdk gphone64 x86 64 (emulator-5554)
[3]: Windows (windows)
[4]: Chrome (chrome)
[5]: Edge (edge)
Please choose one (or "q" to quit): 1
Launching lib\main.dart on SM A356E in debug mode...
Running Gradle task 'assembleDebug'...                            18.5s
√ Built build\app\outputs\flutter-apk\app-debug.apk
Installing build\app\outputs\flutter-apk\app-debug.apk...          6.5s
D/FlutterJNI(26473): Beginning load of flutter...
D/FlutterJNI(26473): flutter (null) was loaded normally!
I/flutter (26473): [IMPORTANT:flutter/shell/platform/android/android_context_vk_impel
Syncing files to device SM A356E...                                69ms

Flutter run key commands.
r Hot reload.
```

12:07
68%
Personal Details
Name: Bina
Course: BCA
College: Sarvajanik College of Commerce & Computer Application


12:08
68%
Search
Amazon
Home02
Quickhome
Apphome
Quick
Assistant
fitnesss
HD Video D...
GoodShort
Storizz
Pratilipi FM
Pratilipi
STORY TV
Wattpad
IndianOil O...
Pilgrim India
flutter_appli...
flutter_appli...